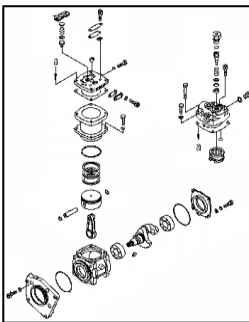
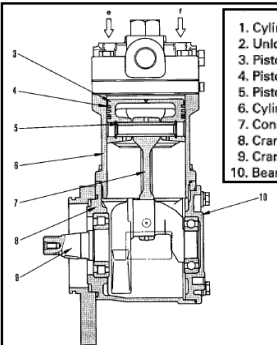


|             |  |
|-------------|--|
| ASSEMBLY BY |  |
|-------------|--|

1 of 5

| No. | PARTS                  | CHECK POINT | CONDITIONS TO BE CHECKED                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SPEC.                             | SIGN |    |    | NOTES, DIAGRAMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                        |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                   | MEC  | QA | GL |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1   | All Parts              |             | Make sure all parts are ready to used<br>(No Scratch and Worn)                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |      |    |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2   | Flange and Crankcase   |             | <p>Fit O-ring to crankcase then assembly Flange tighten with mounting bolts</p> <p>★ Coat thread with thread locker (Three Bond 1360 or LT 2)</p> <p>Fit outer race front bearing to crankcase crankcase</p> <p>★ Coat outer race bearing with bearing lockite or equivalent.</p> <p>Make sure that surface flange must be free from worn and scratch.</p> <p>Install flange front that has a lubrication hole and more thick than rear side of flange<br/>(For 12V140-1 Series)</p> |                                   |      |    |    |  <div data-bbox="1359 902 1489 1070"> <p>1. Cylinder head<br/>2. Unloader valve<br/>3. Piston<br/>4. Piston ring<br/>5. Piston pin<br/>6. Cylinder<br/>7. Connecting rod<br/>8. Crank case<br/>9. Crankshaft<br/>10. Bearing cover</p> </div> <div data-bbox="1145 1247 1398 1357"> <p>a. Air inlet<br/>b. Air outlet<br/>c. Unload pressure (from air governor)<br/>d. Oil inlet<br/>e. Water inlet or outlet<br/>f. Water inlet or outlet</p> </div> |
| 3   | Crankshaft and Con.Rod |             | <p>Fit front bearing to crankshaft smoothly</p> <p>★ Use hydraulic pressure</p> <p>Insert crankshaft in to crankcase, insert big end connecting rod from cylinder mounting fortian then push in crankshaft (clearance between bearing outer race is zero or less)</p> <p>Make sure the position of connecting rod is correct<br/>Press fit ball bearing inner to Crankshaft</p> <p>★ Use hydraulic pressure</p>                                                                      | <p>± 100 Psi</p> <p>± 100 Psi</p> |      |    |    | <p>Be careful when fit of bearing crankshaft.</p> <p>Con rod code (align to front side)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4   | Cover Bearing          |             | <p>Fit O-ring then press fit bearing cover to ball bearing</p> <p>★ Hit the bearing cover with plastic hammer slowly</p> <p>★ Tightening the mounting bolts of the Cover</p> <div data-bbox="504 1895 769 1957"> <p>Actual :                      kgm</p> </div> <p>Spray engine oil bearing and then rotate the crankshaft slowly</p>                                                                                                                                               | 2.5 ~3.0 kgm                      |      |    |    | <p>Make sure that the O-ring is not drop</p> <p>Make sure that the crankshaft rotation is in normal condition</p>                                                                                                                                                                                                                                                                                                                                                                                                                          |

## AIR COMPRESSOR ASSEMBLY CHECK SHEET KOMATSU ENGINE 6D140-2

|               |           |                    |                |              |                  |
|---------------|-----------|--------------------|----------------|--------------|------------------|
| JOB NUMBER    |           | PART NAME          | AIR COMPRESSOR | DOCUMENT NO. | RC/QR04/EG/09/02 |
| ENGINE MODEL  | 6D140-2   | PART NUMBER        |                | PUBL. DATE   | 01/08/2008       |
| ENGINE S/N    |           | START DATE / TIME  |                | REVISION NO. | 01               |
| MACHINE MODEL | GD 825A-2 | FINISH DATE / TIME |                | PAGE         |                  |

| No. | PARTS                                                 | CHECK POINT         | CONDITIONS TO BE CHECKED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SPEC.                                                          | SIGN |    |    | NOTES, DIAGRAMS<br>COMMENTS                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|-------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------|----|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                       |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                | MEC  | QA | GL |                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5   | Piston ring<br>Piston<br>Pump<br>and Cyl.<br>Assembly |                     | <p>Install oil ring and piston rings to piston</p> <p>★ Install Piston Ring with special tool (Ring Expander)</p> <p>Assembly piston rings with the stamp ring surface facing up, and make sure that the end gaps of the rings are in different places and same condition with assembly engine piston ring</p> <p>Oil ring is not top or bottom surface coat cylinder inner with engine oil (SAE 15W ~ 40)</p> <p>★ Install Oil Ring with special tool (Ring Expander)</p> <p>Insert the piston assembly to cylinder liner from top side until to the side and align piston pin hole and small end of connecting rod then push in piston pin and secure bolts end with snap rings</p> <p>★ Piston ring coat with engine oil (SAE15W~40)</p> <p>Fit O-ring to cylinder liner and assembly cylinder to crankcase and tighten with mounting bolts</p> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">             Actual :                      kgm           </div> | <p>End gaps of ring should be facing up</p> <p>2.7~3.3 kgm</p> |      |    |    | <p>Make sure that the ring piston arrangement are in correct position</p> <p>1. Top ring code : R and B</p> <p>2. Second ring code : R</p> <p>Be careful when insert piston assembly to cylinder liner from crack / broken of ring piston</p> <p>Check piston code (    ), does not face plate name or code ( L ) must face plate name</p> <p>Make sure that cylinder liner must be correct specification unit model</p> |
| 6   | Cylinder<br>Head<br>Group                             | Delivery Valve Assy | <p>Install delivery valve, delivery valve guide and stagger then tighten with hexagon socket head screws</p> <p>★ Coat thread with thread locker (Three Bond 1360 or LT 2)</p> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">             Actual :                      kgm           </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0.4~0.5 kgm                                                    |      |    |    | <p>Be careful not to let the delivery valve and delivery valve guide overlap</p> <p>Be careful not to get the thread tighten on the delivery valve</p>                                                                                                                                                                                                                                                                   |
|     |                                                       | Inlet Valve Assy    | <p>Assembly inlet valve seat to cylinder head then tighten mounting bolt</p> <p>Thread and washer must be coat with thread tighten</p> <p>★ Coat thread with thread locker (Three Bond 1360 or LT 2)</p> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">             Actual :                      kgm           </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0.54~0.66 Kgm                                                  |      |    |    | <p>Be careful when install gasket head with liquid must face cylinder head plate</p>                                                                                                                                                                                                                                                                                                                                     |
|     |                                                       | Plate               | <p>Check plate must be free from worn and scratch</p> <p>Tighten plug or stopper valve plate thread must be coat with thread tighten</p> <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">             Actual :                      kgm           </div> <p>Insert washer, inlet valve spray and inlet plate to plate then assembly gasket and O ring and tighten push inlet valve plate from in loader valve end to check movement.</p> <p>Check that the tip of the hexagon socket head screw does not protrude from the surface of plate</p>                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.54~0.66 Kgm                                                  |      |    |    | <p>Tighten with Hexagon socket head screw</p>                                                                                                                                                                                                                                                                                       |



AIR COMPRESSOR  
ASSEMBLY CHECK SHEET  
KOMATSU ENGINE 6D140-2

|               |           |                    |                |              |                  |
|---------------|-----------|--------------------|----------------|--------------|------------------|
| JOB NUMBER    |           | PART NAME          | AIR COMPRESSOR | DOCUMENT NO. | RC/QR04/EG/09/02 |
| ENGINE MODEL  | 6D140-2   | PART NUMBER        |                | PUBL. DATE   | 01/08/2008       |
| ENGINE S/N    |           | START DATE / TIME  |                | REVISION NO. | 01               |
| MACHINE MODEL | GD 825A-2 | FINISH DATE / TIME |                | PAGE         |                  |

## AIR COMPRESSOR ASSEMBLY CHECK SHEET KOMATSU ENGINE 6D140-2

|               |           |                    |                |              |                  |
|---------------|-----------|--------------------|----------------|--------------|------------------|
| JOB NUMBER    |           | PART NAME          | AIR COMPRESSOR | DOCUMENT NO. | RC/QR04/EG/09/02 |
| ENGINE MODEL  | 6D140-2   | PART NUMBER        |                | PUBL. DATE   | 01/08/2008       |
| ENGINE S/N    |           | START DATE / TIME  |                | REVISION NO. | 01               |
| MACHINE MODEL | GD 825A-2 | FINISH DATE / TIME |                | PAGE         |                  |

| No. | PARTS            | CHECK POINT    | CONDITIONS TO BE CHECKED                                                                                                                                                                                                                                                                                                                                             | SPEC.       | SIGN |    |    | NOTES, DIAGRAMS<br>COMMENTS                                                                                                                                                                    |
|-----|------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                  |                |                                                                                                                                                                                                                                                                                                                                                                      |             | MEC  | QA | GL |                                                                                                                                                                                                |
|     |                  | Unloader Valve | Assembly O-ring to rod , insert unloader valve spring and washer then install snap ring<br><br>Install un loader valve body to cylinder head.<br><br>Thread of unloader valve must be coat with thread tighten<br><br>Fit cylinder head gasket to top surface of cylinder head to cylinder then tighten with mounting bolts<br>Tighten the bolts slowly and securely |             |      |    |    |                                                                                                                                                                                                |
| 7   | Coupling and Nut |                | Fit key to groove of crankshaft and install spline shaft, tighten the nut slowly<br><br><div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">             Actual :                      kgm           </div>                                                                                                                   | 25~28 kgm   |      |    |    | Use special tools<br>  |
| 8   | Final Inspection |                | Recheck all bolts from loaser and give recheck marks<br>Turn crankshaft slowly and make sure the crankshaft condition is good.<br><br>For normal condition give stand code (Job No.)<br><br>Playing all holes and pack the compressor<br><br>Check condition un loader valve group are worked                                                                        | White marks |      |    |    |                                                                                                                                                                                                |

### MECHANIC ASSEMBLY COMMENT

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MECH)

QUALITY ASSURANCE ( QA )

GL/SPV PRODUCTION

**AIR COMPRESSOR  
ASSEMBLY CHECK SHEET  
KOMATSU ENGINE 6D140-2**

|               |           |                    |                |              |                  |
|---------------|-----------|--------------------|----------------|--------------|------------------|
| JOB NUMBER    |           | PART NAME          | AIR COMPRESSOR | DOCUMENT NO. | RC/QR04/EG/09/02 |
| ENGINE MODEL  | 6D140-2   | PART NUMBER        |                | PUBL. DATE   | 01/08/2008       |
| ENGINE S/N    |           | START DATE / TIME  |                | REVISION NO. | 01               |
| MACHINE MODEL | GD 825A-2 | FINISH DATE / TIME |                | PAGE         |                  |